

# Course Topics

## Table of contents

|          |                                                                    |          |
|----------|--------------------------------------------------------------------|----------|
| <b>1</b> | <b>BUS 411 — Fixed Income</b>                                      | <b>2</b> |
| 1.1      | Lecture Topics                                                     | 2        |
| 1.2      | Lecture 1: History and Role of Fixed Income Markets                | 3        |
| 1.2.1    | 1.1 What is fixed income?                                          | 3        |
| 1.2.2    | 1.2 Historical development of fixed income markets                 | 3        |
| 1.2.3    | 1.3 Economic role of fixed income securities                       | 3        |
| 1.3      | Lecture 2: Time Value of Money and Discounting                     | 3        |
| 1.3.1    | 2.1 Time value of money                                            | 3        |
| 1.3.2    | 2.2 Compounding conventions                                        | 3        |
| 1.3.3    | 2.3 Discounting fixed income cash flows                            | 4        |
| 1.4      | Lecture 3: Annuities and Liability Valuation                       | 4        |
| 1.4.1    | 3.1 Fixed annuities                                                | 4        |
| 1.4.2    | 3.2 Variable annuities                                             | 4        |
| 1.4.3    | 3.3 Defined benefit pension plans                                  | 4        |
| 1.4.4    | 3.4 Liability valuation                                            | 4        |
| 1.5      | Lecture 4: Government Bonds                                        | 5        |
| 1.5.1    | 4.1 Types of government securities                                 | 5        |
| 1.5.2    | 4.2 Features of government bonds                                   | 5        |
| 1.5.3    | 4.3 Valuation of government bonds                                  | 5        |
| 1.5.4    | 4.4 Government bond markets                                        | 5        |
| 1.6      | Lecture 5: Corporate Bonds and Credit Risk                         | 5        |
| 1.6.1    | 5.1 Corporate bond features                                        | 5        |
| 1.6.2    | 5.2 Valuation of corporate bonds                                   | 6        |
| 1.6.3    | 5.3 Credit risk                                                    | 6        |
| 1.6.4    | 5.4 Corporate bond market considerations                           | 6        |
| 1.7      | Lecture 6: Yield Measures and the Term Structure of Interest Rates | 6        |
| 1.7.1    | 6.1 Yield measures                                                 | 6        |
| 1.7.2    | 6.2 Spot and forward rates                                         | 6        |
| 1.7.3    | 6.3 Yield curve construction and interpretation                    | 6        |

|        |                                                                        |    |
|--------|------------------------------------------------------------------------|----|
| 1.7.4  | 6.4 Term structure concepts . . . . .                                  | 7  |
| 1.8    | Lecture 7: Duration . . . . .                                          | 7  |
| 1.8.1  | 7.1 Interest rate risk in fixed income . . . . .                       | 7  |
| 1.8.2  | 7.2 Macaulay duration . . . . .                                        | 7  |
| 1.8.3  | 7.3 Modified duration . . . . .                                        | 7  |
| 1.8.4  | 7.4 Determinants of duration . . . . .                                 | 7  |
| 1.9    | Lecture 8: Convexity . . . . .                                         | 8  |
| 1.9.1  | 8.1 Limits of duration-only approximation . . . . .                    | 8  |
| 1.9.2  | 8.2 Convexity measure . . . . .                                        | 8  |
| 1.9.3  | 8.3 Duration-convexity approximation . . . . .                         | 8  |
| 1.9.4  | 8.4 Comparing bonds using convexity . . . . .                          | 8  |
| 1.10   | Lecture 9: Fixed Income Risk Management . . . . .                      | 8  |
| 1.10.1 | 9.1 Duration-based risk management . . . . .                           | 8  |
| 1.10.2 | 9.2 Immunization . . . . .                                             | 8  |
| 1.10.3 | 9.3 Convexity in risk management . . . . .                             | 9  |
| 1.10.4 | 9.4 Practical limitations . . . . .                                    | 9  |
| 1.11   | Lecture 10: Bonds with Contingencies . . . . .                         | 9  |
| 1.11.1 | 10.1 Bonds with embedded options . . . . .                             | 9  |
| 1.11.2 | 10.2 Valuation issues for contingent bonds . . . . .                   | 9  |
| 1.11.3 | 10.3 Mortgage-backed and prepayable securities . . . . .               | 9  |
| 1.11.4 | 10.4 Risk characteristics . . . . .                                    | 10 |
| 1.12   | Lecture 11: Fixed Income Derivative Securities . . . . .               | 10 |
| 1.12.1 | 11.1 Overview of fixed income derivatives . . . . .                    | 10 |
| 1.12.2 | 11.2 Interest rate forwards and futures . . . . .                      | 10 |
| 1.12.3 | 11.3 Interest rate swaps . . . . .                                     | 10 |
| 1.12.4 | 11.4 Options on interest rates . . . . .                               | 10 |
| 1.13   | Coverage of Core Course Themes . . . . .                               | 10 |
| 1.13.1 | 1. History of fixed income securities . . . . .                        | 11 |
| 1.13.2 | 2. Valuation methods for fixed income securities . . . . .             | 11 |
| 1.13.3 | 3. Fixed income risk management using duration and convexity . . . . . | 11 |

## 1 BUS 411 — Fixed Income

### 1.1 Lecture Topics

This document lists the planned lecture topics and subtopics for the course.

## **1.2 Lecture 1: History and Role of Fixed Income Markets**

### **1.2.1 1.1 What is fixed income?**

- Definition of fixed income securities
- Debt versus equity financing
- Main issuers and investors

### **1.2.2 1.2 Historical development of fixed income markets**

- Early sovereign borrowing
- Development of government debt markets
- Growth of corporate bond markets

### **1.2.3 1.3 Economic role of fixed income securities**

- Funding governments
  - Funding corporations
  - Saving, retirement, and liability management
- 

## **1.3 Lecture 2: Time Value of Money and Discounting**

### **1.3.1 2.1 Time value of money**

- Present value
- Future value
- Discount factors

### **1.3.2 2.2 Compounding conventions**

- Annual compounding
- Semiannual compounding
- Continuous compounding

### **1.3.3 2.3 Discounting fixed income cash flows**

- Cash flow timelines
  - Pricing as present value of future cash flows
  - Relation between discount rates and value
- 

## **1.4 Lecture 3: Annuities and Liability Valuation**

### **1.4.1 3.1 Fixed annuities**

- Ordinary annuities
- Annuities due
- Perpetuities

### **1.4.2 3.2 Variable annuities**

- Basic structure
- Investment-linked cash flows
- Valuation considerations

### **1.4.3 3.3 Defined benefit pension plans**

- Nature of pension liabilities
- Discounting promised payments
- Funding and valuation issues

### **1.4.4 3.4 Liability valuation**

- Present value of liabilities
  - Matching assets to liabilities
  - Liability sensitivity to interest rates
-

## **1.5 Lecture 4: Government Bonds**

### **1.5.1 4.1 Types of government securities**

- Treasury bills
- Treasury notes
- Treasury bonds

### **1.5.2 4.2 Features of government bonds**

- Face value
- Coupon rate
- Maturity
- Clean price and dirty price

### **1.5.3 4.3 Valuation of government bonds**

- Pricing from discounted cash flows
- Yield to maturity
- Accrued interest

### **1.5.4 4.4 Government bond markets**

- Primary market
  - Secondary market
  - Benchmark role of sovereign debt
- 

## **1.6 Lecture 5: Corporate Bonds and Credit Risk**

### **1.6.1 5.1 Corporate bond features**

- Secured and unsecured bonds
- Seniority
- Covenants

### **1.6.2 5.2 Valuation of corporate bonds**

- Pricing with promised cash flows
- Credit spread over government rates
- Yield spread interpretation

### **1.6.3 5.3 Credit risk**

- Default risk
- Recovery risk
- Credit ratings

### **1.6.4 5.4 Corporate bond market considerations**

- Liquidity
  - Issuer quality
  - Term to maturity and risk
- 

## **1.7 Lecture 6: Yield Measures and the Term Structure of Interest Rates**

### **1.7.1 6.1 Yield measures**

- Current yield
- Yield to maturity
- Holding period return

### **1.7.2 6.2 Spot and forward rates**

- Spot rates
- Forward rates
- Discount function

### **1.7.3 6.3 Yield curve construction and interpretation**

- Normal, flat, and inverted curves
- Economic interpretation
- Expectations about future rates

#### **1.7.4 6.4 Term structure concepts**

- Expectations view
  - Liquidity preference
  - Market segmentation intuition
- 

### **1.8 Lecture 7: Duration**

#### **1.8.1 7.1 Interest rate risk in fixed income**

- Bond price and yield relationship
- Why prices fall when yields rise

#### **1.8.2 7.2 Macaulay duration**

- Definition
- Weighted average time to cash flow receipt
- Interpretation

#### **1.8.3 7.3 Modified duration**

- Definition
- Approximate price sensitivity
- Duration as first-order risk measure

#### **1.8.4 7.4 Determinants of duration**

- Coupon
  - Maturity
  - Yield level
-

## **1.9 Lecture 8: Convexity**

### **1.9.1 8.1 Limits of duration-only approximation**

- Nonlinearity in bond prices
- Curvature of the price-yield relationship

### **1.9.2 8.2 Convexity measure**

- Definition
- Interpretation
- Why convexity matters

### **1.9.3 8.3 Duration-convexity approximation**

- First-order and second-order effects
- Improved bond price change estimates
- Application to larger yield changes

### **1.9.4 8.4 Comparing bonds using convexity**

- Price sensitivity differences
  - Trade-off between yield and convexity
  - Portfolio implications
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## **1.10 Lecture 9: Fixed Income Risk Management**

### **1.10.1 9.1 Duration-based risk management**

- Measuring portfolio interest rate exposure
- Portfolio duration

### **1.10.2 9.2 Immunization**

- Asset-liability matching
- Duration matching
- Protecting against small parallel shifts in yields

### **1.10.3 9.3 Convexity in risk management**

- Why duration matching alone is not enough
- Role of convexity in immunization
- Rebalancing issues

### **1.10.4 9.4 Practical limitations**

- Non-parallel yield curve shifts
  - Model risk
  - Transaction costs
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## **1.11 Lecture 10: Bonds with Contingencies**

### **1.11.1 10.1 Bonds with embedded options**

- Callable bonds
- Puttable bonds
- Convertible features as a general idea

### **1.11.2 10.2 Valuation issues for contingent bonds**

- Cash flows depend on future events
- Option-like features in bond valuation
- Impact on price and yield

### **1.11.3 10.3 Mortgage-backed and prepayable securities**

- Prepayment risk
- Negative convexity
- Valuation challenges

#### **1.11.4 10.4 Risk characteristics**

- Reinvestment risk
  - Call risk
  - Extension and contraction risk
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### **1.12 Lecture 11: Fixed Income Derivative Securities**

#### **1.12.1 11.1 Overview of fixed income derivatives**

- Why derivatives are used
- Hedging, speculation, and portfolio management

#### **1.12.2 11.2 Interest rate forwards and futures**

- Basic contract structure
- Hedging interest rate exposure
- Relation to underlying fixed income securities

#### **1.12.3 11.3 Interest rate swaps**

- Fixed-for-floating exchange
- Basic valuation intuition
- Uses in risk management

#### **1.12.4 11.4 Options on interest rates**

- Caps
  - Floors
  - Basic role in managing rate risk
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### **1.13 Coverage of Core Course Themes**

The course is organized around three major themes:

### **1.13.1 1. History of fixed income securities**

- Covered primarily in Lecture 1

### **1.13.2 2. Valuation methods for fixed income securities**

- Fixed and variable annuities: Lecture 3
- Government bonds: Lecture 4
- Corporate bonds: Lecture 5
- Defined benefit pension plans: Lecture 3
- Bonds with contingencies: Lecture 10
- Fixed income derivative securities: Lecture 11

### **1.13.3 3. Fixed income risk management using duration and convexity**

- Duration: Lecture 7
- Convexity: Lecture 8
- Risk management and immunization: Lecture 9